

Sai Sri Harsha Kumbam

Test Engineer

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SUMMARY

Results-driven **Software Test Engineer** with expertise in **automating complex workflows, hardware/software integration, and production testing** in high-volume manufacturing and enterprise environments. Skilled in **Python, Bash, Ansible, Selenium, TestNG, Docker, PXE, CI/CD pipelines, MySQL**, and in **developing automation tools and dashboards** to improve system reliability, reduce manual effort, and optimize production efficiency. Experienced in **API testing, performance/load testing, and configuration-driven automation frameworks**, collaborating with **cross-functional teams (R&D, Quality, Operations, DevOps)** to streamline processes, troubleshoot critical issues, and deliver high-quality products.

PROFESSIONAL EXPERIENCE

Test Engineer | Quanta Computer Inc. 廣達電腦, TN, USA

05/27/2025- Present

- Developed and maintained **test automation frameworks using Python and Bash scripting**, automating operational workflows to reduce manual effort, improve test coverage, and minimize production defects in high-volume environments.
- Built, configured, and validated **server infrastructure for cloud and AI workloads**, ensuring scalability, reliability, and alignment with client requirements.
- Implemented **configuration management and deployment automation using Ansible**, enabling consistent, scalable updates across multiple racks and reducing deployment time.
- Designed and enhanced a **real-time monitoring and data processing platform**, integrating hardware test stations, PXE systems, and databases to enable end-to-end visibility and operational insights.
- Migrated applications to a **containerized architecture using Docker and NGINX**, implementing load balancing to improve system scalability, availability, and fault tolerance.
- Developed full-stack solutions using **React, Node.js/Express, REST APIs, MySQL, and JSON-based configuration frameworks**, enabling dynamic workflows, real-time monitoring, and data visualization for hardware test operations.
- Implemented **logging, monitoring, and error-handling frameworks**, and performed **root cause analysis (RCA)** to resolve hardware and software integration issues, reducing downtime and improving system reliability.
- Automated **hardware diagnostics and log collection (BMC/BIOS, firmware-level debugging)**, accelerating issue resolution and improving cross-team efficiency.
- Analyzed large-scale production data using **SQL**, improving test capacity, yield, and overall manufacturing performance, utilized **Git for version control and CI/CD integration**, ensuring collaboration, traceability, and efficient code management.

Software Engineer | GVA Infotech , VA, USA

03/03/2025 – 05/23/2025

- Developed and executed **comprehensive test plans and automated test cases** for functional, regression, performance, and integration testing across cloud-based and enterprise applications.
- Automated **UI and API tests** using Selenium, JUnit/TestNG, and Cucumber, significantly reducing manual effort and improving overall test coverage.
- Designed and implemented **automated testing frameworks** using Java, Python, and Groovy, enabling continuous quality assurance within CI/CD pipelines leveraging Jenkins and GitHub Actions.
- Performed extensive **API and microservices testing** using REST interfaces, Postman, and containerized services on Docker and Kubernetes, validating system integration and reliability.
- Collaborated with cross-functional teams including development, product management, and DevOps to troubleshoot defects, perform root cause analysis, and streamline the software delivery process.
- Executed **performance and load testing** using JMeter and LoadRunner, identifying bottlenecks and improving stability for cloud-native applications on AWS and Azure.
- Ensured **end-to-end quality assurance** in Agile/Scrum environments, contributing to scalable and maintainable software platforms across automotive and healthcare domains.

Programmer Analyst | Cognizant, Hyderabad, India

08/17/2021 – 01/13/2023

- Developed **end-to-end automation for regression testing** using Selenium, TestNG, and Maven, reducing manual effort and

accelerating web application deployments.

- Designed and executed **API and microservices test scenarios** in Postman, validating RESTful services and ensuring robust system integration during Agile sprints.
- Authored and maintained **test cases in TestLink**, improving coverage for fintech applications and supporting end-to-end validation of critical payment workflows.
- Managed the **defect lifecycle in Jira**, collaborating closely with development and QA teams to resolve issues efficiently and maintain software quality.
- Built a **data-driven testing framework using Apache POI and TestNG**, automating complex scenarios by dynamically reading/writing Excel test data.
- Partnered with **cross-functional teams** including developers, product managers, and DevOps engineers to troubleshoot defects, perform root cause analysis, and streamline testing processes.
- Contributed to **continuous integration efforts** by integrating automated tests into CI/CD pipelines using Jenkins and Git, ensuring faster feedback and higher release confidence.
- Supported **knowledge sharing and process improvements**, mentoring junior testers and documenting reusable test strategies for team efficiency.
- Ensured **compliance and quality standards** were maintained for enterprise and fintech applications, aligning with both internal QA processes and client expectations.

Software Testing Intern | Cognizant, Hyderabad, India

03/09/2021 – 08/13/2021

- Developed and maintained **Selenium automation scripts** for multiple projects, reducing manual testing effort and improving test efficiency.
- Collaborated with development teams to **identify and resolve defects early**, preventing post-launch issues.
- Generated **reports and dashboards using Excel and Jira**, ensuring timely updates for stakeholders and improving project tracking.
- Assisted in **functional, regression, and integration testing**, gaining experience across multiple application layers.
- Supported **test case design and execution**, contributing to improved test coverage and overall software quality.

SKILLS

- **Automation & Scripting:** Python, Bash, Ansible, Selenium WebDriver, TestNG, PyTest, Maven, Apache POI, Cucumber (BDD), Cypress, Data-Driven & Hybrid frameworks, Page Object Model (POM)
- **Full-Stack & Monitoring:** React (class components), Node.js/Express, MySQL, JSON-driven configuration, real-time monitoring dashboards
- **API & Integration Testing:** Postman, RESTful services, REST Assured, gRPC microservices
- **CI/CD & DevOps:** Jenkins, GitHub Actions, Git, Azure DevOps, Docker, NGINX, PXE boot environments, server provisioning
- **Manual Testing & QA Processes:** Test case design, STLC, cross-browser testing (Chrome, Firefox, Safari), mobile testing (iOS/Android)
- **Test Management & Defect Tracking:** Jira, TestLink, TestRail, Zephyr, Mantis BT
- **Databases & Data Analysis:** SQL, MySQL (querying, validation, production/test data analysis)
- **Performance Testing:** JMeter, LoadRunner
- **Cloud & Infrastructure:** AWS, Azure
- **Methodologies & Practices:** Agile/Scrum, Sprint Planning, Retrospectives, CI/CD integration, cross-functional collaboration
- **Version Control:** Git, GitHub/GitLab

CERTIFICATIONS

- Oracle Certified Associate, Java SE 8 Programmer
- **Cognizant Academy** - Test Case Creation – Simulation, Effective Test Case Creation, Test Data Preparation
- Certified Jr. Test Analyst/Quality Engineer – Selenium

- Test Analyst/Quality Engineer – JavaScript

EDUCATIONS

Masters in Information Systems

01/30/2023 – 12/23/2024

University of Maryland Baltimore County (UMBC), Baltimore, United States | GPA: 3.70/4.00

Bachelor of Technology in Electrical and Electronics Engineering

08/07/2017 – 06/18/2021

Geethanjali College of Engineering and Technology, Hyderabad, India | GPA: 3.67/4.00

